

Unit 1 : Set Theory & Relations

Set Theory& Relations: Introduction, Combination of sets. Relations: Definition, Operations on relations, Properties of relations, Composite Relations, Equality of relations, Recursive definition of relation, Order of relations.

POSET & Lattices: Hasse Diagram, POSET, Definition & Properties of lattices – Bounded, Complemented, Distributed, Modular and Complete lattice.

1. Set Theory

◇ Definition of Set

- A **set** is a well-defined collection of distinct objects or elements.
- The elements can be **numbers, alphabets, symbols, or real-life objects**.
- Sets are usually denoted by **capital letters (A, B, C, ...)** and elements by **small letters (a, b, c, ...)**.

■ Example:

$A = \{1, 2, 3, 4, 5\}$ — here A is a set of first five natural numbers.

◇ Ways of Representing a Set

1. Roster Form (Tabular Form):

- a. All elements are listed within curly brackets $\{\}$.
- b. Example: $A = \{2, 4, 6, 8\}$

2. Set Builder Form:

- a. Describes the property that defines elements of the set.
- b. Example: $A = \{x \mid x \text{ is an even number less than } 10\}$

◇ Types of Sets

- **Finite Set:** Has a limited number of elements.

👉 Example: $\{1, 2, 3, 4\}$

- **Infinite Set:** Has unlimited elements.

👉 Example: $\{1, 2, 3, \dots\}$

- **Null (Empty) Set ($\emptyset$):** Contains no element.
- **Singleton Set:** Contains exactly one element.

- **Equal Sets:** Two sets with same elements.
- **Subset:** A is a subset of B if all elements of A are in B.

👉 Notation: $A \subseteq B$

- **Proper Subset:** $A \subset B$ means $A \subseteq B$ and $A \neq B$
- **Universal Set (U):** Contains all elements under discussion.
- **Power Set (P(A)):** Set of all subsets of A.
- **Complement of a Set (A'):** Elements not in A but in the universal set U.

◇ *Combination of Sets (Set Operations)*

1. Union ($A \cup B$):

All elements that are in A or in B or in both.

👉 Example: $A = \{1, 2, 3\}$, $B = \{3, 4, 5\} \Rightarrow A \cup B = \{1, 2, 3, 4, 5\}$

2. Intersection ($A \cap B$):

Common elements in both sets.

👉 Example: $A \cap B = \{3\}$

3. Difference ($A - B$):

Elements in A but not in B.

👉 Example: $A - B = \{1, 2\}$

4. Complement (A'):

Elements in universal set but not in A.

5. Symmetric Difference:

$$(A \Delta B) = (A - B) \cup (B - A)$$

◇ *Venn Diagrams*

- Used to represent sets and their relationships graphically.
- Circles represent sets inside a rectangle (universal set).
- Helpful to visualize operations like union, intersection, and complement.

🌸 2. Relations

📌 Definition of Relation

- A **relation** shows how elements of one set are **connected or associated** with elements of another set.
- If A and B are two sets, a **relation R from A to B** is a subset of the **Cartesian product $A \times B$** .

■ Example:

Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$

$A \times B = \{(1,4), (1,5), (1,6), (2,4), (2,5), (2,6), (3,4), (3,5), (3,6)\}$

If $R = \{(1,4), (2,5), (3,6)\}$, then R is a relation from A to B.

📌 Operations on Relations

1. Inverse Relation (R^{-1}):

If $(a, b) \in R$, then $(b, a) \in R^{-1}$.

👉 Example: If $R = \{(1,2), (2,3)\} \Rightarrow R^{-1} = \{(2,1), (3,2)\}$

2. Complement of a Relation:

All pairs in $A \times B$ that are not in R.

3. Union / Intersection / Difference of Relations:

Same as for sets, applied on ordered pairs.

📌 Properties of Relations

1. Reflexive:

$(a, a) \in R$ for all a in A.

👉 Example: $R = \{(1,1), (2,2), (3,3)\}$ is reflexive.

2. Symmetric:

If $(a, b) \in R \Rightarrow (b, a) \in R$.

👉 Example: $R = \{(1,2), (2,1)\}$

3. Transitive:

If $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.

4. Antisymmetric:

If $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$.

Comparison Table: Properties of Relations

Property	Meaning (in simple words)	Condition	Example (A = {1, 2, 3})	Key Point
Reflexive	Every element is related to itself.	$(a, a) \in R$ for all $a \in A$	$R = \{(1,1), (2,2), (3,3), (1,2)\}$	All elements have self-loops.
Irreflexive	No element is related to itself.	$(a, a) \notin R$ for all $a \in A$	$R = \{(1,2), (2,3)\}$	No element has a self-loop.
Non-Reflexive	Some elements are related to themselves, some are not.	$(a, a) \in R$ for some a , not for all	$R = \{(1,1), (2,3)\}$	Partial reflexivity — mix of both.

Property	Meaning (simple)	Condition	Example (A = {1,2,3})	Hint to Identify
Symmetric	If a is related to b, then b is also related to a.	$(a,b) \in R \Rightarrow (b,a) \in R$	$R = \{(1,2), (2,1)\}$	Two-way relation.
Anti-symmetric	If a is related to b and b to a, then $a = b$.	$(a,b) \in R$ and $(b,a) \in R \Rightarrow a = b$	$R = \{(1,1), (2,2), (1,2)\}$	Both directions only for same element.
Asymmetric	If a is related to b, then b is not related to a.	$(a,b) \in R \Rightarrow (b,a) \notin R$	$R = \{(1,2), (2,3)\}$	Only one direction, never both.

Property	Meaning (in simple words)	Condition	Example	Extra Info
Transitive	If a is related to b and b to c, then a is related to c.	$(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$	$R = \{(1,2), (2,3), (1,3)\}$	Forms a logical "chain".
Equivalence Relation	A relation that is Reflexive + Symmetric + Transitive .	Must satisfy all three properties together.	"aRb if a - b is divisible by 3"	Divides a set into equivalence classes .

◊ Composite Relation

- If R is a relation from $A \rightarrow B$ and S from $B \rightarrow C$,

then $S \circ R$ (read “S composite R”) is a relation from $A \rightarrow C$.

- Defined as:

$$S \circ R = \{(a, c) \mid \exists b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}$$

 Example:

$$A = \{1, 2\}, B = \{3, 4\}, C = \{5, 6\}$$

$$R = \{(1, 3), (2, 4)\}, S = \{(3, 5), (4, 6)\}$$

$$\Rightarrow S \circ R = \{(1, 5), (2, 6)\}$$

◇ **Equality of Relations**

- Two relations R_1 and R_2 are **equal** if they contain exactly the same ordered pairs.

👉 i.e., $R_1 = R_2 \Leftrightarrow$ every pair in R_1 is in R_2 and vice versa.

◇ **1. Function (Basic)**

- A **function** is a rule that gives **one output for each input**.
- Example: $f(x) = 2x + 3$
 - If $x = 2 \rightarrow f(2) = 7$

◇ **2. Composite Function**

When one function’s output becomes input of another function.

👉 **Definition:**

If we have two functions f and g :

- **fog(x)** = $f(g(x)) \rightarrow g$ first, then f
- **gof(x)** = $g(f(x)) \rightarrow f$ first, then g

 Example 1:

$$f(x) = 2x + 3, \quad g(x) = x^2$$

- $\text{fog}(x) = f(g(x)) = f(x^2) = 2x^2 + 3$
- $\text{gof}(x) = g(f(x)) = g(2x + 3) = (2x + 3)^2 = 4x^2 + 12x + 9$

✓ $\text{fog}(x) \neq \text{gof}(x)$

Example 2:

$$f(x) = x + 1, \quad g(x) = 3x$$

- $f \circ g(x) = 3x + 1$
- $g \circ f(x) = 3x + 3$

Example 3:

$$f(x) = x^2, \quad g(x) = \sqrt{x}$$

- $f \circ g(x) = f(\sqrt{x}) = (\sqrt{x})^2 = x$
- $g \circ f(x) = g(x^2) = \sqrt{x^2} = |x|$

✓ $f \circ g(x) = x$, $g \circ f(x) = |x| \rightarrow$ both not always same.

◇ 3. Inverse Function (f^{-1})

Inverse function **reverses** the effect of the original function.

If $f(x) = y$, then $f^{-1}(y) = x$

So,

- $f(f^{-1}(x)) = x$
- $f^{-1}(f(x)) = x$

Example 4:

$$f(x) = 2x + 3$$

Find $f^{-1}(x)$

Steps:

1. $y = 2x + 3$
2. Swap x and $y \rightarrow x = 2y + 3$
3. Solve for $y \rightarrow y = (x - 3)/2$ ✓ $f^{-1}(x) = (x - 3)/2$

Example 5:

$$f(x) = 3x - 2$$

$$f^{-1}(x) = (x + 2)/3$$

Then,

- $f(f^{-1}(x)) = 3((x + 2)/3) - 2 = x$

$$\bullet \quad f^{-1}(f(x)) = ((3x - 2) + 2)/3 = x$$

✓ Both give $x \rightarrow$ Correct inverse!

◇ 4. Composite with Inverse

Expression	Meaning	Result
$f(f^{-1}(x))$	Inverse first, then f	x
$f^{-1}(f(x))$	f first, then inverse	x

Both are **Identity Functions** (result = x).

◇ 5. Important Points to Remember

- In **fog**, apply **g first**, then **f**.
- In **gof**, apply **f first**, then **g**.
- A function has inverse **only if it is one-one and onto**.
- For all inverse functions:

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

✓ Quick Summary Table

Concept	Formula	Meaning
$fog(x)$	$f(g(x))$	g first, then f
$gof(x)$	$g(f(x))$	f first, then g
$f^{-1}(x)$	Inverse of $f(x)$	Reverses function
$f(f^{-1}(x))$	Identity function	$= x$
$f^{-1}(f(x))$	Identity function	$= x$

◇ Recursive Definition of Relation

- A **recursive relation** defines elements of a relation in terms of **previous elements**.
- Common in **mathematical sequences or recurrence relations**.

■ Example:

The relation R on natural numbers defined as:

- Base case: $(1, 1) \in R$
- Recursive step: If $(a, b) \in R$, then $(a+1, b+2) \in R$

Then $R = \{(1, 1), (2, 3), (3, 5), (4, 7) \dots\}$

◇ Order of Relation

- The **order** of a relation refers to the **number of elements (ordered pairs)** it contains.
- Example:

If $R = \{(1, 2), (2, 3), (3, 4)\}$,

then order of $R = 3$

☑ Extra Topics (Useful for Exam)

1. Domain of Relation:

Set of all first elements of ordered pairs.

2. Range of Relation:

Set of all second elements of ordered pairs.

3. Equivalence Relation:

A relation that is **Reflexive, Symmetric, and Transitive**.

Example: "is equal to" relation ($=$)

4. Partial Order Relation:

A relation that is **Reflexive, Antisymmetric, and Transitive**.

PART 2: POSET & Lattices

✿ 1. POSET (Partially Ordered Set)

◇ Definition

- A **partially ordered set (POSET)** is a set P together with a **binary relation ($\leq$)** that satisfies:
 - **Reflexive:** $a \leq a$
 - **Antisymmetric:** if $a \leq b$ and $b \leq a \Rightarrow a = b$

- **Transitive:** if $a \leq b$ and $b \leq c \Rightarrow a \leq c$
- The relation “ $\leq$ ” defines **partial ordering** on set P.

■ Example:

Let $P = \{1, 2, 3, 4, 6\}$

Define relation $a \leq b$ if a divides b .

Then $(P, \leq)$ is a **POSET** because it satisfies all three properties.

◇ Terminology

- **Comparable elements:**
 a, b are comparable if either $a \leq b$ or $b \leq a$.
- **Incomparable elements:**
 a and b are incomparable if neither $a \leq b$ nor $b \leq a$.
- **Minimal element:**
 No other element is smaller than it.
- **Maximal element:**
 No other element is greater than it.
- **Least element:**
 Smaller than or equal to all other elements.
- **Greatest element:**
 Greater than or equal to all other elements.

■ Example:

In $P = \{1, 2, 3, 6\}$ with “divides” relation,

- Minimal element $\rightarrow 1$
- Maximal element $\rightarrow 6$

✿ 2. Hasse Diagram

◇ Definition

- A **Hasse diagram** is a **graphical representation** of a **POSET** that shows the partial order relation **without drawing reflexive and transitive edges**.

◇ Rules for Drawing

1. Represent each element as a **node** (point).
2. Draw an **edge upward** from a to b if $a < b$ and **no element c** exists such that $a < c < b$ (i.e., immediate relation).
3. **Do not** include self-loops or transitive links.

4. The **lowest element(s)** are placed at the **bottom** and **greatest** at the **top**.

■ **Example:**

For $(P, \leq)$ where $P = \{1, 2, 4, 8\}$ and $a \leq b$ if a divides b ,

The Hasse diagram looks like:



This shows how each element divides the next.

✿ 3. Lattices

◇ Definition

- A **lattice** is a **POSET** in which **every pair of elements** has both:
 - a **Least Upper Bound (LUB)** or **join ($\vee$)**
 - a **Greatest Lower Bound (GLB)** or **meet ($\wedge$)**

■ Example:

Consider the set **$P = \{1, 2, 4, 8\}$** with relation “divides”.

- $\text{LUB}(2, 4) = 4$
- $\text{GLB}(2, 4) = 2$

Hence, $(P, \text{divides})$ forms a **lattice**.

◇ Important Terms

1. Join ($\vee$):

- a. Least upper bound of two elements.
- b. Example: $2 \vee 4 = 4$

2. Meet ($\wedge$):

- a. Greatest lower bound of two elements.
- b. Example: $2 \wedge 4 = 2$

◇ *Properties of Lattices*

1. Idempotent Law:

a. $a \vee a = a$ and $a \wedge a = a$

2. Commutative Law:

a. $a \vee b = b \vee a$ and $a \wedge b = b \wedge a$

3. Associative Law:

a. $(a \vee b) \vee c = a \vee (b \vee c)$

b. $(a \wedge b) \wedge c = a \wedge (b \wedge c)$

4. Absorption Law:

a. $a \vee (a \wedge b) = a$

b. $a \wedge (a \vee b) = a$

✿ 4. Types of Lattices

◇ (a) *Bounded Lattice*

- A lattice is **bounded** if it has:
 - a **least element (0)** — smallest in the set
 - a **greatest element (1)** — largest in the set

■ Example:

For divisibility lattice $\{1, 2, 4, 8\}$,

- Least element (0) = 1
- Greatest element (1) = 8

◇ (b) *Complemented Lattice*

- A bounded lattice is **complemented** if every element **a** has a **complement a'** such that:
 - $a \vee a' = 1$
 - $a \wedge a' = 0$

■ Example:

In Boolean algebra lattice,

- $a = 0 \Rightarrow a' = 1$

- $a = 1 \Rightarrow a' = 0$

◇ (c) *Distributive Lattice*

- A lattice is **distributive** if it satisfies **distributive laws**:
 - $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
 - $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

■ Example:

The Boolean algebra $\{0, 1\}$ with $\vee$ and $\wedge$ operations is distributive.

● Note:

Not every lattice is distributive (e.g., diamond-shaped lattices are non-distributive).

◇ (d) *Modular Lattice*

- A lattice $(L, \vee, \wedge)$ is **modular** if for all a, b, c in L :
 - If $a \leq c$, then $a \vee (b \wedge c) = (a \vee b) \wedge c$

■ Example:

The divisibility lattice of positive integers is **modular**, but not always distributive.

◇ (e) *Complete Lattice*

- A lattice is **complete** if **every subset** (not just pairs) has both a **join (LUB)** and **meet (GLB)**.
- In other words, even for large sets of elements, we can find LUB and GLB.

■ Example:

The power set of any set $(P(A), \subseteq)$ is a **complete lattice** because union = LUB and intersection = GLB.

UNIT 1 – Important Theory Questions & Answers

- ◆ 1. Define a Set. Explain its different types with examples.

A **set** is a well-defined collection of distinct objects or elements.

It is usually denoted by capital letters (A, B, C, ...).

Types of Sets:

1. **Finite Set:** Has limited number of elements.
2. Example: $A = \{1, 2, 3, 4\}$
3. **Infinite Set:** Unlimited elements.

Example: $N = \{1, 2, 3, \dots\}$

4. **Empty Set ($\emptyset$):** Contains no elements.
5. **Singleton Set:** Has only one element.
6. **Equal Sets:** Two sets having same elements.
7. **Subset:** $A \subseteq B$ means every element of A is in B.
8. **Proper Subset:** $A \subset B$ means $A \subseteq B$ and $A \neq B$.
9. **Universal Set:** Contains all elements under discussion.
10. **Power Set:** Set of all subsets of a set.
11. **Complement:** Elements not in the given set.

2. Explain the operations on sets with examples.

The main operations on sets are:

Operation	Symbol	Description	Example
Union	$A \cup B$	All elements in A or B or both	$A = \{1, 2\}, B = \{2, 3\} \Rightarrow A \cup B = \{1, 2, 3\}$
Intersection	$A \cap B$	Common elements	$A \cap B = \{2\}$
Difference	$A - B$	Elements in A but not in B	$A - B = \{1\}$
Complement	A'	Elements not in A (from Universal set)	
Symmetric Difference	$A \Delta B$	Elements in A or B but not both	$A \Delta B = \{1, 3\}$

3. What is a Venn Diagram? Explain with an example.

A **Venn diagram** is a **pictorial representation of sets** using circles.

Each circle represents a set, and overlapping regions represent common elements.

Example:

- $A \cup B \rightarrow$ total area covered by both circles.
- $A \cap B \rightarrow$ common overlapping area.

Venn diagrams help in visualizing relationships like **union, intersection, and complement**.

◆ **4. Define Cartesian Product of Sets with example.**

The **Cartesian product** of two sets A and B, written as $A \times B$, is the set of all **ordered pairs (a, b)** where $a \in A$ and $b \in B$.

■ **Example:**

If $A = \{1, 2\}$, $B = \{x, y\}$,

Then $A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$.

The number of elements = $|A| \times |B| = 4$.

◆ **5. Define Relation and give examples.**

A **relation R** from set A to set B is a subset of the **Cartesian product $A \times B$** .

It shows how elements of one set are connected with elements of another set.

■ **Example:**

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$

$R = \{(1, 4), (2, 5), (3, 6)\}$

Here, R is a relation from A to B.

◆ **6. Explain the Properties of Relations.**

Let R be a relation on set A. Then:

1. Reflexive:

$(a, a) \in R$ for every $a \in A$

Example: “ $\leq$ ” on numbers.

2. Symmetric:

$(a, b) \in R \Rightarrow (b, a) \in R$

Example: “is brother of”

3. Transitive:

$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$

Example: “is greater than”

4. Antisymmetric:

$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$

Example: “ $\leq$ ” relation on numbers.

◆ **7. What is a Composite Relation?**

If R is a relation from $A \rightarrow B$ and S is a relation from $B \rightarrow C$,
then $S \circ R$ (S composite R) is a relation from $A \rightarrow C$ defined as:

$$\text{👉 } S \circ R = \{(a, c) \mid \exists b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}$$

■ **Example:**

$$A = \{1, 2\}, B = \{3, 4\}, C = \{5, 6\}$$

$$R = \{(1, 3), (2, 4)\}, S = \{(3, 5), (4, 6)\}$$

$$\text{Then } S \circ R = \{(1, 5), (2, 6)\}$$

◆ **8. Define Inverse of a Relation.**

If R is a relation from A to B , then its **inverse relation** (R^{-1}) is the relation from B to A defined as:

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

■ **Example:**

$$R = \{(1, 2), (2, 3)\}$$

$$R^{-1} = \{(2, 1), (3, 2)\}$$

◆ **9. What is an Equivalence Relation?**

A relation R on a set A is an **equivalence relation** if it is:

1. **Reflexive**
2. **Symmetric**
3. **Transitive**

■ **Example:**

On set of integers Z ,
relation R defined as **aRb if $(a - b)$ is divisible by 5**
is an equivalence relation.

◆ **10. What is Recursive Definition of Relation?**

A **recursive relation** defines elements using **previous elements** of the relation.

■ **Example:**

Define relation R on natural numbers:

- Base case: $(1, 1) \in R$
- Recursive step: If $(a, b) \in R \Rightarrow (a+1, b+2) \in R$

$$\text{Then } R = \{(1, 1), (2, 3), (3, 5), (4, 7), \dots\}$$

◆ **11. What is the Order of a Relation?**

The **order of a relation** is the **number of ordered pairs** it contains.

If $R = \{(1,2), (2,3), (3,4)\}$, then order of $R = 3$.

◆ **12. What are Domain, Range, and Co-domain of a Relation?**

- **Domain:** Set of all first elements.
- **Range:** Set of all second elements that actually appear.
- **Co-domain:** Set of possible second elements (B).

■ Example:

If $R = \{(1,2), (2,3), (3,4)\}$ from $A = \{1,2,3\}$ to $B = \{2,3,4,5\}$,

- Domain = $\{1,2,3\}$
- Range = $\{2,3,4\}$
- Co-domain = $\{2,3,4,5\}$

◆ **13. Differentiate between Partial Order and Equivalence Relation.**

Feature	Partial Order	Equivalence Relation
Properties	Reflexive, Antisymmetric , Transitive	Reflexive, Symmetric , Transitive
Example	" $\leq$ " on numbers	"is equal to" on numbers
Structure	Used in POSETs	Used in partitioning sets

◆ **14. Write short note on Equality of Relations.**

Two relations R_1 and R_2 are **equal** if they contain exactly the same ordered pairs.

That is,

$R_1 = R_2 \Leftrightarrow (a, b) \in R_1 \Rightarrow (a, b) \in R_2$ and vice versa.